

Theorem .0.1

Let L/K be a finite Galois extension of number fields with Galois group

$$G = \text{Gal}(L/K),$$

and let $\mathfrak{p} \subseteq \mathcal{O}_K$ be a nonzero prime ideal. For an intermediate field

$$K \subseteq M \subseteq L,$$

write $Q_M = Q \cap \mathcal{O}_M$, which is a prime ideal. Then

Then:

(1)

$$e(Q_M/\mathfrak{p}) = 1 \iff M \subseteq L^I,$$

where $I = I(Q/\mathfrak{p})$ is the inertia group.

(2)

$$e(Q_M/\mathfrak{p}) = f(Q_M/\mathfrak{p}) = 1 \iff M \subseteq L^D,$$

where $D = D(Q/\mathfrak{p})$ is the decomposition group.

Moreover,

$$Q_{L^I} \cdot \mathcal{O}_L = Q^{e(Q/P)}$$

and

$$Q_{L^D} \mathcal{O}_{L^I} = \mathfrak{q}_1 \cdots \mathfrak{q}_g$$

is a product of distinct prime ideals, each having the same residue field degree over Q_{L^D}

Proof. We prove (2). We first show that $e(Q_{L^D}/\mathfrak{p}) = f(Q_{L^D}/\mathfrak{p}) = 1$. Let $L^D = K'$, and $\mathfrak{p}' = \mathfrak{p} \cap \mathcal{O}_{K'}$, $D' = D(Q/\mathfrak{p}')$, $I' = D \cap I = I$. By the lemma, $D' = D \cap D = D$. By HW problem,

$$e(Q_i/\mathfrak{p}) = e(Q_i/\mathfrak{p}')e(\mathfrak{p}'/\mathfrak{p})$$

and

$$f(Q_i/\mathfrak{p}) = f(Q_i/\mathfrak{p}')f(\mathfrak{p}'/\mathfrak{p})$$

since $D' = D$, $I' = I$. We get $e(Q_{L^D}/\mathfrak{p}) = f(Q_{L^D}/\mathfrak{p}) = 1$. Now suppose $K \hookrightarrow K' \hookrightarrow L$ with $e(\mathfrak{p}'/\mathfrak{p}) = 1$ and $f(\mathfrak{p}'/\mathfrak{p}) = 1$. Let $K' = L^H$ for some subgroup H of G . Then $I' = I \cap H$ and $D' = D \cap H$. So

$$L^{D'} = L^{D \cap H} = L^D \cdot L^H = L^D \cdot K'$$

Using the assumption that $e(\mathfrak{p}'/\mathfrak{p}) = f(\mathfrak{p}'/\mathfrak{p}) = 1$, we get

$$\begin{aligned} e(Q/\mathfrak{p}) &= e(Q/\mathfrak{p}') \cdot e(\mathfrak{p}'/\mathfrak{p}) \\ &= e(Q/\mathfrak{p}') \end{aligned}$$

and

$$\begin{aligned} f(Q/\mathfrak{p}) &= f(Q/\mathfrak{p}') \cdot f(\mathfrak{p}'/\mathfrak{p}) \\ &= f(Q/\mathfrak{p}') \end{aligned}$$

so $|I| = |I'|$ and $|D| = ef = |D'|$. Therefore, $I' = I$, $D' = D$, and $K' \subseteq L^D$.

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Proposition .0.2

Let M_1/K and M_2/K be extensions of number fields, and let

$$\mathfrak{p}$$

be a nonzero prime ideal of $\mathcal{O}_K$.

- (1) Suppose $\mathfrak{p}$ is unramified in both M_1/K and M_2/K ; that is, for every prime ideal

$$\mathfrak{q}_j \subseteq \mathcal{O}_{M_j} \quad (j = 1, 2)$$

lying above $\mathfrak{p}$, one has

$$e(\mathfrak{q}_j/\mathfrak{p}) = 1.$$

Then $\mathfrak{p}$ is unramified in the compositum M_1M_2/K .

Conversely, if $\mathfrak{p}$ is unramified in M_1M_2/K , then it is unramified in both M_1/K and M_2/K .

Hence

$$\mathfrak{p} \text{ is unramified in } M_1/K \text{ and } M_2/K \iff \mathfrak{p} \text{ is unramified in } M_1M_2/K.$$

- (2) Assume $\mathfrak{p}$ splits completely in both M_1/K and M_2/K ; i.e.

$$\mathfrak{p}\mathcal{O}_{M_j} = \prod_{i=1}^{[M_j:K]} \mathfrak{q}_{j,i} \quad (j = 1, 2),$$

where the $\mathfrak{q}_{j,i}$ are distinct prime ideals of $\mathcal{O}_{M_j}$.

Then $\mathfrak{p}$ splits completely in M_1M_2/K .

Conversely, if $\mathfrak{p}$ splits completely in M_1M_2/K , then it splits completely in both M_1/K and M_2/K .

Therefore

$$\mathfrak{p} \text{ splits completely in } M_1/K \text{ and } M_2/K \iff \mathfrak{p} \text{ splits completely in } M_1M_2/K.$$

Proof. We first prove (1).

If $\mathfrak{p}$ is unramified in M_1M_2/K , let Q be any prime ideal of $\mathcal{O}_{M_1M_2}$ lying above $\mathfrak{p}$, and let

$$R = Q \cap \mathcal{O}_{M_1}.$$

Then by multiplicativity of ramification indices,

$$1 = e(Q/\mathfrak{p}) = e(Q/R) e(R/\mathfrak{p}).$$

Hence

$$e(R/\mathfrak{p}) = 1,$$

so $\mathfrak{p}$ is unramified in M_1/K . Similarly, $\mathfrak{p}$ is unramified in M_2/K .

Conversely, suppose $\mathfrak{p}$ is unramified in both M_1/K and M_2/K . Let

$$L/K$$

be a finite Galois extension containing M_1M_2 , and let Q be a prime ideal of $\mathcal{O}_L$ lying above $\mathfrak{p}$. Let

$$I = I(Q/\mathfrak{p})$$

be the inertia group. Since $\mathfrak{p}$ is unramified in M_1/K , the remark gives

$$M_1 \subseteq L^I.$$

Likewise,

$$M_2 \subseteq L^I.$$

Therefore

$$M_1M_2 \subseteq L^I.$$

Applying the remark again, we conclude that

$$e(Q_{M_1M_2}/\mathfrak{p}) = 1.$$

Since this holds for every prime of M_1M_2 above $\mathfrak{p}$, the prime $\mathfrak{p}$ is unramified in M_1M_2/K .

This proves (1).

For (2), the argument is similar. If $\mathfrak{p}$ splits completely in M_1M_2/K , then for every prime $Q_{M_1M_2}$ of M_1M_2 above $\mathfrak{p}$,

$$e(Q_{M_1M_2}/\mathfrak{p}) = f(Q_{M_1M_2}/\mathfrak{p}) = 1.$$

Passing to subextensions shows that the same holds in M_1 and M_2 , so $\mathfrak{p}$ splits completely there as well.

Conversely, suppose $\mathfrak{p}$ splits completely in both M_1/K and M_2/K . Let L/K be a finite Galois extension containing M_1M_2 , let Q_L be a prime of $\mathcal{O}_L$ above $\mathfrak{p}$, and let

$$D = D(Q_L/\mathfrak{p})$$

be the decomposition group. Since $\mathfrak{p}$ splits completely in M_1/K , the remark implies

$$M_1 \subseteq L^D.$$

Similarly,

$$M_2 \subseteq L^D.$$

Hence

$$M_1M_2 \subseteq L^D.$$

Again by the remark,

$$e(Q_{M_1M_2}/\mathfrak{p}) = f(Q_{M_1M_2}/\mathfrak{p}) = 1.$$

Thus $\mathfrak{p}$ splits completely in M_1M_2/K .

This proves (2).

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